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MATHEMATICS

STANDARD FOUR

National Assessment 2022
Answer Key and Worked Solutions

Subject	Mathematics
Code	04E
Level	Standard Four
Year	2022
Exam Board	NECTA
Exam	National Assessment
Type	Answer key and worked solutions
Questions	5 questions, each with 5 parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S4-2022.pdf

Question 1: Numbers and Place Value

1(a). Ana eats $\frac{1}{4}$ of an orange. What fraction remains?

Step 1: One whole orange is $\frac{4}{4}$.

Step 2: $\frac{4}{4} - \frac{1}{4} = \frac{3}{4}$.

Final Answer: $\frac{3}{4}$.

1(b). Write 99,001 in expanded form.

Step 1: 9 ten-thousands = 90,000; 9 thousands = 9,000; 1 one = 1.

Step 2: The hundreds and tens places are zero.

Final Answer: $90,000 + 9,000 + 1$.

1(c). Find the value of X in XVIII.

Step 1: In a Roman numeral, X stands for ten.

Step 2: XVIII = $10 + 5 + 3$.

Final Answer: 10.

1(d). Write the number between 48 and 50 in Roman numerals.

Step 1: The number is 49.

Step 2: $40 = XL$ and $9 = IX$; combine them.

Final Answer: XLIX.

1(e). What number must be added to 4,392 to make 7,005?

Step 1: Unknown addend = total - known addend.

Step 2: $7,005 - 4,392 = 2,613$.

Step 3: Check: $4,392 + 2,613 = 7,005$.

Final Answer: 2,613.

Section Summary: Read each digit's place and distinguish ordinary numbers from Roman numerals.

Question 2: Sequences and Comparison

2(a). A tortoise moves 5 cm in one step. How many steps cover 30 cm?

Step 1: Divide the total distance by the distance per step.

Step 2: $30 \div 5 = 6$.

Final Answer: 6 steps.

2(b). Five children differ in age by 5 years. The first-born is 35. Find the fourth child's age.

Step 1: Move three places from the first to the fourth child.

Step 2: $35 - 3 \times 5 = 20$.

Final Answer: 20 years.

2(c). Using the years given in the paper (1962, 1961, 1963 and 1980), which country gained independence first?

Step 1: The earliest year in the supplied data is 1961.

Step 2: That year is paired with Tanganyika.

Final Answer: Tanganyika.

Study Tip: This is a comparison of the supplied numbers; use the earliest year in the question's table.

2(d). Gladness gets 186 marks in month 1 and improves by 12 each month. Find month 4's marks.

Step 1: There are three increases from month 1 to month 4.

Step 2: $3 \times 12 = 36$; $186 + 36 = 222$.

Final Answer: 222 marks.

2(e). School elections occur every two years. If the last was 2018, when is the next?

Step 1: Add the two-year interval to 2018.

Step 2: $2018 + 2 = 2020$.

Final Answer: 2020.

Section Summary: Find the repeated change first; count the gaps between terms.

Question 3: Operations and Time

3(a). Find the difference between 39,365 and 33,868.

Step 1: Subtract the smaller number from the larger.

Step 2: $39,365 - 33,000 = 6,365$; $6,365 - 868 = 5,497$.

Final Answer: 5,497.

3(b). Share 350 mangoes equally into ten groups.

Step 1: Equal sharing means division.

Step 2: $350 / 10 = 35$.

Final Answer: 35 mangoes per group.

3(c). Baraka earns sh. 4,000 daily for 40 days. Find his pay.

Step 1: Total pay = daily pay x days.

Step 2: $4,000 \times 40 = 160,000$.

Final Answer: Sh. 160,000.

3(d). A radio sells for sh. 60,500 with profit sh. 20,000. Find its buying price.

Step 1: Profit = selling price - buying price.

Step 2: Buying price = $60,500 - 20,000$.

Final Answer: Sh. 40,500.

3(e). Convert 21 days to weeks.

Step 1: One week has seven days.

Step 2: $21 \div 7 = 3$.

Final Answer: 3 weeks.

Section Summary: Keep the units consistent. Regroup every 60 minutes as one hour.

Question 4: Shapes and Measurement

4(a). Arrange 900 m, 2 cm, 3 km and 400 mm in descending order.

Step 1: Convert to metres: 900 m, 0.02 m, 3,000 m, 0.4 m.

Step 2: Order $3,000 > 900 > 0.4 > 0.02$.

Final Answer: 3 km, 900 m, 400 mm, 2 cm.

4(b). Which is longer, 20 cm of wire or 2 m of thread?

Step 1: Convert 2 m to 200 cm.

Step 2: 200 cm is greater than 20 cm.

Final Answer: The 2 m thread.

4(c). How many line segments are determined by the four points W, X, Y and Z?

Step 1: From W: WX, WY, WZ (three).

Step 2: From X, count new segments XY, XZ (two).

Step 3: From Y, count YZ (one); total $3 + 2 + 1$.

Final Answer: 6 line segments.

Study Tip: Count every distinct pair of endpoints, including segments that span other marked points.

4(d). Find the perimeter of the triangle with sides 13 m, 14 m and 15 m.

Step 1: Perimeter means the total distance around the shape.

Step 2: $13 + 14 + 15 = 42$.

Final Answer: 42 m.

4(e). A square has perimeter 196 cm. Find one side.

Step 1: There are four equal sides.

Step 2: $196 / 4 = 49$.

Final Answer: 49 cm.

Section Summary: Use the shape's sides, perimeter formula and metric units carefully.

Question 5: Reading Data

5(a). The pupil counts are 2014:80, 2015:90, 2016:95, 2017:98. Which year has the most?

Step 1: Compare the counts 80, 90, 95, 98.

Step 2: 98 is greatest and belongs to 2017.

Final Answer: 2017.

5(b). Find the pupils completing school in 2014 and 2015 together.

Step 1: Read 2014 = 80 and 2015 = 90.

Step 2: $80 + 90 = 170$.

Final Answer: 170 pupils.

5(c). Which year has the fewest completing pupils?

Step 1: The smallest count is 80.

Step 2: The table pairs 80 with 2014.

Final Answer: 2014.

5(d). Find the difference between the 2014 and 2017 counts.

Step 1: Read 2014 = 80 and 2017 = 98.

Step 2: $98 - 80 = 18$.

Final Answer: 18 pupils.

5(e). Find the total completing pupils in all four years.

Step 1: $80 + 90 = 170$.

Step 2: $95 + 98 = 193$.

Step 3: $170 + 193 = 363$.

Final Answer: 363 pupils.

Section Summary: Read the key, count every symbol, and distinguish the year from the numerical value.